SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – C Code Output Prediction

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	**Assessment:**	Assessment Tool 1 – C Code Output Prediction
Weightage:	35% of CIA	**Total Marks:**	100
CO1 Statement:	*Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.*		
Student Name:	Siva Charan	**Reg. No.:**	192525039
Section / Batch:	AI and ML/ 2025	**Date:**	10/07/2026

Code of Conduct

I Siva Charan ______________________________(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Siva Charan __________

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1 - 2	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	3 - 4	20
Search Data Structure	Linear Search and Binary Search	5 -6	20
Recursion, Performance analysis	Recursion	7 - 8	20
Array Data Structures	Implementations Using Array data structure	9 - 10	20
		GRAND TOTAL:	**100 Marks**

Q1 **C Code – Array Traversal**

```
#include <stdio.h>
int main() {
int arr[] = {5, 10, 15, 20};
int i, sum = 0;
for(i = 0; i < 4; i++)
sum += arr[i];
printf("%d", sum);
return 0;
}
```

Your Answer

50

Q2 **C Code – Linear Search**

```
#include <stdio.h>
int main() {
int arr[] = {2,4,6,8,10};
int key = 8;
int i;
for(i=0;i<5;i++){
if(arr[i]==key){
printf("%d",i);
break;
} } }
```

Your Answer

3

Q3 C Code – Binary Search Mid

```
#include<stdio.h>
int main(){
int low=0,high=7;
int mid;
mid=(low+high)/2;
printf("%d",mid);
}
```

Your Answer

3

Q4 C Code – Pointer with Array

```
#include<stdio.h>
int main(){
int arr[]={10,20,30};
int *p=arr;
printf("%d",*(p+2));
}
```

Your Answer

30

Q5 **C Code – Array Update**

```
#include<stdio.h>
int main(){
int arr[]={1,2,3};
arr[1]=arr[0]+arr[2];
printf("%d",arr[1]);
}
```

Your Answer

4

Q6 C Code - Linked List Node

```
#include<stdio.h>
struct Node{
int data;
struct Node *next;
};
int main(){
struct Node n1,n2;
n1.data=10;
n2.data=20;
n1.next=&n2;
n2.next=NULL;
printf("%d",n1.next->data);
}
```

Your Answer

20

Q7 C Code - Time Complexity Loop

```
for(i=0;i<n;i++)
printf("*");
```

The complexity is

Your Answer

0(n)

Q8 C Code - Nested Loop

```
for(i=0;i<n;i++)
for(j=0;j<n;j++)
printf("*");
```

Your Answer

if n=3

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Q9 C Code - Binary Search Complexity

```
while(low<=high){
mid=(low+high)/2;
}
```

Your Answer

0(log n)

Q10 C Code - Pointer Increment

```
#include<stdio.h>
int main(){
int a[]={5,10,15};
int *p=a;
p++;
printf("%d",*p);
}
```

Your Answer

10